New upper bounds for $A(18, 6, 7)$, $A(19, 6, 7)$, and $A(19, 6, 8)$

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Abstract

We prove $A(18, 6, 7) \leq 309$, $A(19, 6, 7) \leq 460$, and $A(19, 6, 8) \leq 667$, improving the previous upper bounds 312, 463, and 673, respectively. We augment Polak's four-point semidefinite bound with anticode vacancy, integer occupancy, and cardinality constraints obtained by established moment methods, and certify the resulting bounds by exact rational dual certificates.

1 Introduction

Let $A(n, d, w)$ denote the largest cardinality of a binary code of length n whose words have weight w and whose distinct words have Hamming distance at least d. Brouwer's tables [1] collect bounds for this function. For $(n, d, w) = (18, 6, 7), (19, 6, 7), (19, 6, 8)$, the previous integer upper bounds are 312, 463, and 673, respectively. The first two are due to Polak [4]; the third was established by Chang [12] using an exact certificate for Polak's four-point bound.

Semidefinite programming bounds for codes include Schrijver's three-point bound based on the Terwilliger algebra [2], the quadruple-distance bound of Gijswijt, Mittelmann, and Schrijver [3], and Polak's four-point bounds for constant-weight codes [4]. Strengthening an established bound by additional valid inequalities also has precedents: Agrell, Vardy, and Zeger [5] augmented linear programming inequalities for constant-weight codes, and Kim and Toan [6] added linear constraints to Schrijver's semidefinite program.

Here eight families of valid constraints strengthen Polak's bound at the three parameters above. The constructions use Boolean multipliers and linearization in the Sherali–Adams approach [7], localizing matrices [8], finite-support and binomial moment inequalities [9], symmetry reduction [10, 4], and classical intersecting families as anticodes [11]. These are established techniques; their application here gives the three improved bounds and the accompanying exact certificates.

2 The four-point bound and normalization

Write $[n] = \{1, \ldots, n\}$ and identify a word with its support in

$$\mathcal{F} = \binom{[n]}{w}, \qquad V = |\mathcal{F}| = \binom{n}{w}, \qquad d_H(u, v) = 2(w - |u \cap v|).$$

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The relations indexed by $w - |u \cap v|$ form the Johnson scheme. Put $\delta = \lceil d/2 \rceil$. A set of words is *compatible* if its distinct members have mutual distance at least d.

For a code C, put $X_u(C) = \mathbf{1}_{\{u \in C\}}$ and $X_S = \prod_{u \in S} X_u$, with $X_\varnothing = 1$. Boolean reduction uses $X_u^2 = X_u$ and $X_u X_v = 0$ for incompatible distinct words. Thus $X_S X_T = X_{S \cup T}$, with incompatible unions interpreted as zero. For a uniformly distributed coordinate permutation $g \in S_n$, define

$$z_C(S) = V \mathbb{E}_g[X_S(gC)], \qquad |S| \le 4. \tag{1}$$

These moments depend only on the orbit of the unordered set S. They satisfy

$$z_C(\varnothing) = V, \qquad z_C(\{u\}) = |C|, \qquad 0 \le z_C(S) \le V.$$

Write z_1 for the common singleton moment. For a Boolean-reduced polynomial of degree at most four, set

$$\mathcal{L}_z\Big(\sum_S a_S X_S\Big) = \sum_S a_S z(S).$$

In particular, $\mathcal{L}_z(1) = V$ and $\mathcal{L}_{z_C}(p) = V \mathbb{E}_g[p(gC)]$. The affine constant of the semidefinite program is 1; throughout the paper $z(\varnothing)$ denotes the fixed number V. A term $a z(\varnothing)$ therefore contributes the affine constant aV.

For an anchor $D \subseteq \mathcal{F}$, define

$$\mathcal{I}_D = \{S \subseteq \mathcal{F} : D \subseteq S,\ |D| + 2|S \setminus D| \le 4\}, \qquad M_D(z)_{S,T} = z(S \cup T).$$

The underlying four-point program has the constraints

$$\begin{aligned} M_\varnothing(z) \succeq 0, \qquad M_D(z) \succeq 0 \quad (D \text{ compatible},\ |D| = 2), \\ z(S) \ge 0 \quad (S \text{ compatible},\ 1 \le |S| \le 4), \end{aligned} \tag{2}$$

together with $z(\varnothing) = V$, orbit invariance, and zero moments for incompatible sets. Maximizing z_1 gives Polak's bound $A_4(n, d, w)$ [4]. In his normalization, $x(S) = z(S)/V$, so $x(\varnothing) = 1$ and $\sum_{u \in \mathcal{F}} x(\{u\}) = z_1$. The original constraints and their upper-bound property are those of [4, Section 1.1]; its Sections 1.2 and 3 give the coordinate averaging and representation block reductions. Section references to [4] refer to the linked arXiv version.

All additional constraints use the following positivity principle. If $p(C) \ge 0$ for every code, then $\mathcal{L}_{z_C}(p) \ge 0$. If $h(C) \ge 0$ and f is a real vector of polynomials, then

$$\mathcal{L}_{z_C}(h f f^{\mathsf{T}}) \succeq 0, \qquad a^{\mathsf{T}} \mathcal{L}_{z_C}(h f f^{\mathsf{T}}) a = V \mathbb{E}_g[h(gC)(a^{\mathsf{T}} f(gC))^2] \ge 0, \tag{3}$$

provided every reduced entry has degree at most four. Fixed congruences $K \mapsto P^{\mathsf{T}} K P$, coordinate averaging, and division by positive common factors preserve positivity. The added matrix families use retained projections, as specified below; their exact rows and blocks are recorded in the released models.

3 Eight additional constraint families

In the last three families, B denotes an integer upper bound on the size of every code with the specified parameters. Section 4 establishes this premise independently from the unaugmented four-point program.

3.1 Anticode vacancy scalars

Put $t = w - \delta + 1$. For $t \geq 0$, $0 \leq r \leq \delta - 1$, and a coordinate set R of feasible size $t + 2r$, let

$$Q(R, r) = \{u \in \mathcal{F} : |u \cap R| \geq t + r\}, \qquad H_Q = 1 - \sum_{q \in Q} X_q.$$

These are the classical intersecting families in [11, eq. (1.9)]. For the present parameters, $\delta = 3$ and $r \in \{0, 1, 2\}$. For each retained compatible three-word anchor D, the added scalar inequality is

$$z(D) - \sum_{q \in Q} z(D \cup \{q\}) \geq 0. \tag{4}$$

Proof. For $u, v \in Q$, $|u \cap v| \geq |u \cap R| + |v \cap R| - |R| \geq t$, so $d_H(u, v) \leq 2(\delta - 1) < d$. Hence $|C \cap Q| \leq 1$ and $H_Q(C) \in \{0, 1\}$ for every code. The left side of (4) at z_C is $\mathcal{L}_{z_C}(X_D H_Q) \geq 0$ and uses at most four distinct words. □

The retained scalar rows have $D \cap Q = \varnothing$ and at least two compatible extension words in Q. Duplicate rows are removed, and positions of R relative to D are taken up to coordinate permutations.

3.2 Anticode vacancy matrices

Fix $Q = Q(R, r)$ and a word a. Let

$$\mathcal{B}_a = \{b \in \mathcal{F} : b = a \text{ or } d_H(a, b) \geq d\}, \qquad K^{a,Q}(z)_{b,c} = z(\{a, b, c\}) - \sum_{q \in Q} z(\{a, b, c, q\}).$$

The constraints are the retained projections

$$P^{\mathsf{T}} K^{a,Q}(z) P \succeq 0. \tag{5}$$

Only harmonic degree zero is used here. More explicitly, the stabilizer of (a, R) permutes the four coordinate cells determined by membership in a and R. If $\mathcal{O}_\beta$ is a word orbit, specified by its four cell weights, the projection has columns $P_{b,\beta} = \mathbf{1}_{\{b \in \mathcal{O}_\beta\}}$. The projected entries are $\sum_{b \in \mathcal{O}_\beta, c \in \mathcal{O}_\gamma} K^{a,Q}(z)_{b,c}$. Degree-zero Young representatives give the same projection up to fixed nonzero column normalizations.

Proof. For $v = (X_b)_{b \in \mathcal{B}_a}$, $K^{a,Q}(z_C) = V \mathbb{E}_g[X_a H_Q v v^{\mathsf{T}}] \succeq 0$ by (3). Its entries have degree at most four, and fixed congruence gives (5). The words b, c are each compatible with a; their entry is zero when they are incompatible with one another. The index $b = a$ supplies the constant coordinate after multiplication by X_a. Rows in Q vanish under H_Q, and the whole matrix vanishes if $a \in Q$; these zero rows and matrices are removed. □

3.3 Matching-region occupancy projectors

Choose $L \subseteq U \subseteq [n]$ and put

$$Q = \{u \in \mathcal{F} : L \subseteq u \subseteq U\}, \qquad m = |U \setminus L|, \qquad k = w - |L|,$$

with $\min(k, m - k) = \delta$ and $3\delta \leq m < 4\delta$. Write $e_0 = 1$ and

$$e_\ell(X_Q) = \sum_{\substack{E \subseteq Q,\ |E| = \ell \\ E \text{ compatible}}} X_E, \qquad P_j = \sum_{\ell = j}^{3} (-1)^{\ell - j} \binom{\ell}{j} e_\ell(X_Q).$$

Thus $P_0 = 1 - e_1 + e_2 - e_3$, $P_1 = e_1 - 2e_2 + 3e_3$, $P_2 = e_2 - 3e_3$, and $P_3 = e_3$. For a word anchor a, the retained constraints are

$$\sum_{\ell=j}^{3} (-1)^{\ell-j} \binom{\ell}{j} \sum_{\substack{E \subseteq Q,\ |E|=\ell \\ E \text{ compatible}}} z(\{a\} \cup E) \geq 0, \qquad j = 0, 1, 2. \tag{6}$$

Proof. Deleting L leaves k-subsets of an m-set. If $k = \delta$, the distance condition makes them disjoint; if $m - k = \delta$, their complements are disjoint. Thus $N = |C \cap Q| \leq \lfloor m/\delta \rfloor = 3$, and $e_\ell(X_Q)(C) = \binom{N}{\ell}$. For $N \geq j$, binomial inversion gives

$$P_j(C) = \binom{N}{j} \sum_{r=0}^{N-j} (-1)^r \binom{N-j}{r} = \mathbf{1}_{\{N=j\}},$$

and the same identity holds when $N < j$, with both sides zero. Consequently (6) is $\mathcal{L}_{z_C}(X_a P_j) \geq 0$. If $a \in Q$, the summation still allows $a \in E$, and set union merges its repeated occurrence. Every term uses at most four distinct words. □

The $j = 3$ inequality is a sum of nonnegative moments. The retained family omits it, zero rows, duplicates, and other rows having only nonnegative coefficients, and takes region and anchor representatives under coordinate permutations.

3.4 Coordinate interval spectra

First define an integer shortening bound, for $0 \leq k \leq m$, by

$$J(m, d, k) = \begin{cases} 1, & \min(k, m-k) < \delta, \\ \min\left\{ \left\lfloor \dfrac{mJ(m-1, d, k-1)}{k} \right\rfloor, \left\lfloor \dfrac{mJ(m-1, d, k)}{m-k} \right\rfloor \right\}, & \text{otherwise.} \end{cases} \tag{7}$$

Lemma 1. *Every length-m, weight-k code of minimum distance at least d has cardinality at most $J(m, d, k)$.*

Proof. In the base case, two words have distance at most $2\min(k, m-k) < d$. Otherwise, for a code E, let E_i^1 and E_i^0 denote the words with a one or zero at coordinate i. Deleting that fixed coordinate preserves distances. Induction and incidence counting give

$$k|E| = \sum_i |E_i^1| \leq mJ(m-1, d, k-1), \qquad (m-k)|E| = \sum_i |E_i^0| \leq mJ(m-1, d, k).$$

Taking integer parts proves (7). The cases $k = 0, m$ are base cases. □

For disjoint coordinate sets L, Z with $|L| = \ell$ and $|Z| = b$, define

$$Q(L, Z) = \{u \in \mathcal{F} : L \subseteq u,\ u \cap Z = \varnothing\}, \qquad N_{L,Z} = \sum_{u \in Q(L,Z)} X_u.$$

The occupancy on a code is an integer in $\{0, \ldots, c\}$, where $c = J(n - \ell - b, d, w - \ell)$. Only feasible intervals with $2 \leq c \leq 20$ are selected. For a selected degree $1 \leq q \leq \min(4, c)$, choose a q-element set $\mathcal{R} \subseteq \{0, \ldots, c\}$ and $s \in \{1, -1\}$ such that

$$p(x) = s \prod_{r \in \mathcal{R}} (x - r) \geq 0 \quad (x = 0, \ldots, c), \qquad p(x) = \sum_{r=0}^{q} a_r \binom{x}{r}, \quad a_r = \Delta^r p(0), \tag{8}$$

where $\Delta p(x) = p(x+1) - p(x)$. The coefficients a_r are integers. The finite-support polynomial description is justified in Appendix A.

Let Ω_r be a set of representatives of the orbits of compatible unordered r-sets of words, and write

$$o(T) = |\operatorname{orb}(T)|, \qquad I(T) = \left|\bigcap_{u\in T} u\right|, \qquad U(T) = \left|\bigcup_{u\in T} u\right|, \qquad H_{\ell,b} = \binom{n}{\ell}\binom{n-\ell}{b}.$$

The retained constraints are

$$a_0 H_{\ell,b} V + \sum_{r=1}^{q} a_r \sum_{T\in\Omega_r} o(T)\binom{I(T)}{\ell}\binom{n-U(T)}{b} z(T) \geq 0. \tag{9}$$

An impossible binomial coefficient is zero.

Proof. For each (L, Z), Lemma 1 gives $p(N_{L,Z}(C)) \geq 0$. Boolean reduction represents $\binom{N_{L,Z}}{r}$ as the sum of X_T over compatible r-subsets of $Q(L, Z)$. Apply $\mathcal{L}_{z_C}$ and sum over all $H_{\ell,b}$ pairs. A nonempty T lies in an interval exactly when L lies in its common intersection and Z lies outside its union. This gives $\binom{I(T)}{\ell}\binom{n-U(T)}{b}$ choices. Grouping by orbit proves (9), including the constant $a_0 H_{\ell,b} V$. Its moments have order at most $q \leq 4$. □

3.5 Coordinate occupancy matrices

Use either orientation $\varepsilon \in \{0, 1\}$, with

$$u^{(0)} = u, \qquad u^{(1)} = [n] \setminus u, \qquad w_0 = w, \quad w_1 = n - w.$$

Complementation preserves distance, and the indicator of $u^{(1)}$ in the complemented code is X_u. For a nonempty core R, $|R| = \ell \leq w_\varepsilon$, put

$$N_R = \sum_{u: R\subseteq u^{(\varepsilon)}} X_u, \qquad F_{R,i} = \sum_{u:|u^{(\varepsilon)}\cap R|=i} X_u, \qquad v_R = (1, (F_{R,i})_i)^{\mathsf{T}},$$

where $\max(0, \ell + w_\varepsilon - n) \leq i \leq \min(\ell, w_\varepsilon)$. Lemma 1 gives $0 \leq N_R(C) \leq c$, with $c = J(n-\ell, d, w_\varepsilon - \ell)$. The selected pairs (ℓ, c) are listed in Table 1. For each pair, use

$$p_j(x) = (x - j)(x - j - 1) \quad (0 \leq j < c), \qquad p_{\text{end}}(x) = x(c - x),$$

and impose

$$\mathcal{O}_{\varepsilon,\ell,p}(z) = \sum_{\substack{R\subseteq[n]\\|R|=\ell}} \mathcal{L}_z\big(p(N_R) v_R v_R^{\mathsf{T}}\big) \succeq 0. \tag{10}$$

Table 1: Selected core sizes and capacities for coordinate occupancy matrices.

(n, d, w)	Original orientation	Complement orientation	Matrices
$(18, 6, 7)$	$(3, 15), (4, 4)$	$(6, 19), (7, 8), (8, 3)$	54
$(19, 6, 7)$	$(3, 20), (4, 5)$	$(7, 19), (8, 8), (9, 3)$	60
$(19, 6, 8)$	$(4, 15), (5, 4)$	$(7, 9), (8, 3)$	35

Proof. For an integer N_R, the consecutive factors $N_R - j$ and $N_R - j - 1$ have nonnegative product. The capacity gives $N_R(c - N_R) \geq 0$. Every summand in (10) is therefore positive semidefinite at z_C by (3), and its reduced degree is at most $2 + 1 + 1 = 4$. Coordinate invariance gives the exact normalization

$$\begin{aligned} \mathcal{O}_{\varepsilon,\ell,p}(z_C) &= V \sum_{|R|=\ell} p(N_R(C)) v_R(C) v_R(C)^{\mathsf{T}} \\ &= V \binom{n}{\ell} \mathbb{E}_R[p(N_R(C)) v_R(C) v_R(C)^{\mathsf{T}}]. \end{aligned} \tag{11}$$

Each selected pair contributes $c + 1$ matrices. The vector v_R is a fixed linear image of the constant and the individual-word indicators, so this construction is also a fixed congruence of a localizing matrix. □

Appendix B gives the orbit coefficients, including all repeated-word terms and both orientations. Zero-row deletion and removal of positive common factors preserve the displayed constraints.

3.6 Global cardinality cap

Let $M(X) = \sum_{u \in \mathcal{F}} X_u$. The independently certified premise $|C| \leq B$ gives $B - M(X) \geq 0$ on every code. The global constraint is

$$B - z_1 \geq 0. \tag{12}$$

Proof. At z_C, the left side is $B - |C| \geq 0$. Its normalization is also given by $V^{-1}\mathcal{L}_z(B - M) = B - V^{-1} \sum_{u \in \mathcal{F}} z(\{u\}) = B - z_1$. Thus the affine constant in this row is B. □

3.7 Anchored cardinality scalars

For each retained nonempty compatible anchor D, $k = |D| \leq 3$, impose

$$Bz(D) - \sum_{u \in \mathcal{F}} z(D \cup \{u\}) = (B - k)z(D) - \sum_{u \in \mathcal{F} \setminus D} z(D \cup \{u\}) \geq 0. \tag{13}$$

Proof. The first expression at z_C is $\mathcal{L}_{z_C}(X_D(B - M)) \geq 0$. The integrand is nonnegative for every code of size at most B, and its reduced degree is at most four. Repeated anchor words give the $kz(D)$ term. □

For compatible orbit representatives $T \in \Omega_k$ and $W \in \Omega_{k+1}$, let

$$m(T, W) = \#\{T' \subseteq W : |T'| = k,\ T' \in \mathrm{orb}(T)\}, \qquad e(T, W) = \frac{o(W)m(T, W)}{o(T)}. \tag{14}$$

The exact orbit form of (13) is

$$(B - k)z(T) - \sum_{W \in \Omega_{k+1}} e(T, W)z(W) \geq 0. \tag{15}$$

The integer incidence count $e(T, W)$ and the treatment of repeated words are proved in Appendix B.2.

3.8 Rooted cardinality matrices

For a word a and $\mathcal{B}_a$ as above, define

$$K^{a,B}(z)_{b,c} = Bz(\{a,b,c\}) - \sum_{u\in\mathcal{F}} z(\{a,b,c,u\}), \qquad b,c \in \mathcal{B}_a.$$

The added constraints are the retained Young projections

$$P^{\mathsf{T}} K^{a,B}(z) P \succeq 0. \tag{16}$$

Proof. For $v = (X_b)_{b\in\mathcal{B}_a}$,

$$K^{a,B}(z_C) = V\mathbb{E}_g[X_a(B-M)vv^{\mathsf{T}}] = (B-|C|)\mathcal{L}_{z_C}(X_a vv^{\mathsf{T}}) \succeq 0.$$

All entries have at most four distinct words; repetitions, including $b = a$ or $c = a$, are merged by union. Fixed congruence proves (16) for every code. □

The stabilizer of a is $S_w \times S_{n-w}$. Its projections use tensor products of binary Young representatives of shapes $(w-h,h)$ and $(n-w-j,j)$, where $h, j \geq 0$, $2h \leq w$, and $2j \leq n-w$. The weight index $s = |a \cap b|$ satisfies

$$h \leq s \leq w-h, \qquad j \leq w-s \leq n-w-j, \qquad s = w \text{ or } w-s \geq \delta.$$

This is the singleton-anchor Young construction of [4, Section 3.1], applied to the cardinality localizer. For compatible $T = \{a,b,c\}$ of size k, the corresponding entry is exactly the left side of (15); incompatible entries are zero. This linear substitution commutes with the fixed projections.

4 The independently certified cardinality premise

The three baseline certificates concern only the original Polak program (2). They give

$$U_{\text{base}}(18,6,7) = \frac{1343302937217311918620211354045644495982287584164198 7}{4294967296000000000000000000000000000000000000000000},$$

$$U_{\text{base}}(19,6,7) = \frac{1274550274376814049016394972588387247712471375324415947}{2748779069440000000000000000000000000000000000000000000},$$

$$U_{\text{base}}(19,6,8) = \frac{8413000178831003092549218915831626644003 13}{1250000000000000000000000000000000000000000}.$$

Every code gives a feasible point of that program, so exact weak duality gives $|C| \leq U_{\text{base}}$. The integrality of $|C|$ then gives

$$B = \lfloor U_{\text{base}} \rfloor = \begin{cases} 312, & (n,d,w) = (18,6,7), \\ 463, & (n,d,w) = (19,6,7), \\ 673, & (n,d,w) = (19,6,8). \end{cases}$$

This establishes $B - M(X) \geq 0$ on all codes, including the empty code and codes of size smaller than B, and justifies the last three families in Section 3. The third baseline bound is the certificate bound of [12].

5 Exact dual certificates and verification

The proofs above show that every code induces a feasible point of the augmented program, with objective $z_1 = |C|$. Only moments of at most four distinct words occur. We give the moment bounds needed to control coefficient residuals on the entire semidefinite feasible region.

Lemma 2. *Every feasible point of the invariant four-point program satisfies* $0 \le z(S) \le V$ *for* $|S| \le 4$.

Proof. Nonnegativity is imposed in (2), and incompatible moments are zero. The principal submatrix of $M_\varnothing$ on $\varnothing, \{u\}$ gives

$$\begin{pmatrix} V & z_1 \\ z_1 & z_1 \end{pmatrix} \succeq 0, \qquad z_1^2 \le V z_1,$$

so $z_1 \le V$. For a compatible pair, the submatrix on $\{u\}, \{u, v\}$ gives $z(\{u, v\})^2 \le z_1 z(\{u, v\})$, hence $z(\{u, v\}) \le z_1$. For a compatible triple, use the indices $D, D \cup \{w\}$ in M_D, where $D = \{u, v\}$, to obtain $z(D \cup \{w\})^2 \le z(D) z(D \cup \{w\})$. Finally, the entry between $D \cup \{w\}$ and $D \cup \{x\}$ gives

$$z(D \cup \{w, x\})^2 \le z(D \cup \{w\}) z(D \cup \{x\}) \le V^2.$$

These inequalities also cover zero diagonal entries and establish the claim. □

To state exact weak duality, index the nonempty orbit variables by i, write the objective as $c^\mathsf{T} z = z_1$, and write every primal block as $F_j(z) = F_{j,0} + \sum_i F_{j,i} z_i \succeq 0$; a scalar inequality is a 1×1 block. For rational positive semidefinite dual matrices Y_j, put

$$\beta = \sum_j \langle Y_j, F_{j,0} \rangle, \qquad r_i = c_i + \sum_j \langle Y_j, F_{j,i} \rangle, \qquad \langle Y, F \rangle = \operatorname{tr}(YF).$$

Exact arithmetic gives the identity

$$z_1 = \beta + \sum_i r_i z_i - \sum_j \langle Y_j, F_j(z) \rangle \le \beta + \sum_i r_i z_i. \tag{17}$$

Lemma 2 bounds the residual term by $V \sum_i \max(r_i, 0)$. Thus a recorded rational residual allowance η that bounds this term yields $z_1 \le \beta + \eta$ for every feasible point. The certified upper bounds include this allowance.

Each certificate records the rational dual matrices, exact LDL^T factors, residual compensation, and hashes of the corresponding JSON and SDPA models. Exact verification establishes positive semidefiniteness and the coefficient identities and bounds in (17). The same argument applies to the independent baseline certificates and to the augmented certificates. The dual matrices were obtained by rounding numerical solutions of the semidefinite programs to rational matrices; the bounds rely only on the exact verification described above.

6 Results

Theorem 1. *The exact rational upper bounds in Table 2 hold. In particular,*

$$\boxed{A(18, 6, 7) \le 309, \qquad A(19, 6, 7) \le 460, \qquad A(19, 6, 8) \le 667.}$$

Proof. For each parameter triple, the baseline certificate establishes the cardinality premise of Section 4. Every code moment vector then satisfies all eight added constraint families. Applying the augmented certificate and (17) bounds its objective $|C|$ by the stated rational number. Taking the integer part gives the three assertions. □

Table 2: Exact certified bounds. The decimal expansions are complete.

(n, d, w)	Rational upper bound	Exact decimal	Integer bound
$(18, 6, 7)$	$\frac{3868791767977833258518625959723}{12500000000000000000000000000}$	309.503341438226660681490076777784	309
$(19, 6, 7)$	$\frac{46047385392564732804912812389921}{10000000000000000000000000000}$	460.473853925647328049128123899210	460
$(19, 6, 8)$	$\frac{16683417296961520597884426836631}{25000000000000000000000000000}$	667.336691878460823915377073465240	667

7 Data and code availability

The certificate package is available at

`https://github.com/wille614/cw-code-upper-bounds`

as release v1.0.0. It contains the three baseline and three augmented models, each in JSON and SDPA form, their six rational dual certificates, an input preparation program, and the exact verifier. The 18 compressed assets have published SHA-256 checksums. The baseline certificate for $(19, 6, 8)$ is byte-identical to the published certificate accompanying [12].

Offline verification uses CPython 3.11 or later, its standard library, and enabled assertions. For each parameter triple, the preparation program checks the six asset hashes, decompresses the models, and checks the augmented model's reference to the baseline certificate, its rational bound, and its integer part. Separate verifier invocations establish the baseline premise and the augmented bound. The release documentation gives commands for Windows, Linux, and macOS. The six verifications have integer-bound pairs $(312, 309)$, $(463, 460)$, and $(673, 667)$; successful runs report `PASS` and exit code 0.

For $(18, 6, 7)$, $(19, 6, 7)$, and $(19, 6, 8)$, respectively, the augmented verifications check 46677, 52716, and 116007 blocks, covering 13834807, 15981482, and 66812277 SDPA coefficients. The code is licensed under Apache License 2.0, and the certificates and model data under CC BY 4.0.

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A Finite-support polynomials

For $q \le c$, let $\mathcal{P}_q$ be the cone of real polynomials of degree at most q that are nonnegative at $0, \dots, c$. Evaluation at these points is injective, so the cone is pointed; the strictly positive constant polynomial shows that it has dimension $q + 1$. An extreme ray must have at least q independent active evaluation constraints. A nonzero polynomial of degree at most q has at most q distinct roots. Thus every extreme ray is a positive multiple of a signed product in (8). Conversely, its q distinct roots give q independent equations, so a valid signed product spans a one-dimensional face. Lower-degree valid products are also allowed. The Newton expansion $p(x) = \sum_r \Delta^r p(0) \binom{x}{r}$ gives the coefficients used in (9).

B Orbit coefficients and repeated words

B.1 Coordinate occupancy matrices

Expand an entry of $N_R^p v_R v_R^{\mathsf{T}}$ for $p = 0, 1, 2$. There are at most four ordered word slots: p region slots and zero, one, or two row slots; the constant row contributes no slot. Repetitions are allowed. Every tuple is specified uniquely by its set T of distinct words and a surjection from the slots onto T. Incompatible T contribute zero. This accounts for the diagonal terms in N_R^2 and for coincidences between row and region words.

For a representative T and a fixed surjection, the core R must lie in the intersection of all oriented region words. With no region slot this eligible set is $[n]$. Partition the eligible set into four cells by the two oriented row-word bits, using bit zero for a missing row. If the sizes are $c_{00}, c_{01}, c_{10}, c_{11}$, the number of possible cores is

$$\sum_{(x_{00},x_{01},x_{10},x_{11})} \prod_{a,b\in\{0,1\}} \binom{c_{ab}}{x_{ab}}, \tag{18}$$

where $\sum_{a,b} x_{ab} = \ell$ and, for present row slots, the intersection requirements are $x_{10} + x_{11} = i$ and $x_{01} + x_{11} = j$. Multiplication by $o(T)$, summation over the surjections, and replacement of X_T by $z(T)$ give the coefficient for that orbit. The empty tuple contributes $\binom{n}{\ell}V$. Combining the powers $p = 0, 1, 2$ with the coefficients of the chosen quadratic gives exactly (10).

For the complement orientation, all coordinate bits are reversed before these counts are formed. The moment is still that of the original T, with the same orbit size. Thus both orientations use the same orbit variables.

B.2 Cardinality extension counts

To justify (14), count pairs (T', W') of the prescribed orbit types with $T' \subseteq W'$. There are $o(W)m(T, W)$ such incidences. Transitivity makes the number above each T' equal, so $e(T, W) = o(W)m(T, W)/o(T)$ is an integer. For fixed T, each containing W' corresponds to exactly one compatible new word extending T. Separating the $k = |T|$ repeated words in the sum over all $u \in \mathcal{F}$ gives (15).

For code-induced moments, invariance of $|gC|$ also gives

$$|C|z_C(T) = kz_C(T) + \sum_{W\in\Omega_{k+1}} e(T, W)z_C(W).$$

For an arbitrary semidefinite vector, the expression on the right is the linearized moment $\mathcal{L}_z(MX_T)$. Applying this expression entrywise to $T = \{a, b, c\}$ gives the rooted cardinality matrix before its fixed Young projection. Orbit averaging, deletion of zero or repeated rows, and positive rescaling preserve the validity of all retained constraints.

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